Mathematics GCE O Level 2020 - Paper 2

Duration: 2h30 | Cameroon GCE Board

Duration: two and a half hours. This paper is arranged in two sections, A and B. Answer ALL questions in Sections A and B. Section A: answer ALL 15 questions in the spaces provided; the mark allocation for each question is indicated. Section B: answer ALL four questions; all questions in Section B carry equal marks (15 marks each). You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed.

Exercise 1 (4 marks)

Simplify $\frac{2}{3} + \frac{3}{4} \div \frac{6}{7} - \frac{11}{12}$. *[4 marks]*

Exercise 2 (7 marks)

a. Given the sets $A = \{0 \leq x < 4,\ x \in \mathbb{Z}\}$ and $B = \{-1 \leq x < 2,\ x \in \mathbb{Z}\}$, list the elements of the set A.

b. Given the sets $A = \{0 \leq x < 4,\ x \in \mathbb{Z}\}$ and $B = \{-1 \leq x < 2,\ x \in \mathbb{Z}\}$, list the elements of the set B.

c. Given the sets $A = \{0 \leq x < 4,\ x \in \mathbb{Z}\}$ and $B = \{-1 \leq x < 2,\ x \in \mathbb{Z}\}$, draw a Venn diagram to represent the sets.

Exercise 3 (5 marks)

Complete the logic table in Figure 1.

p	q	$\sim q$	$p \Rightarrow \sim q$
T	T		
T	F		
F	T		
F	F		

Figure 1 [5 marks]

Exercise 4 (6 marks)

a. Solve the inequality $-7 < 5x - 2 \leq 3$.

b. Represent the solution of (a) on a real number line.

Exercise 5 (7 marks)

The functions g and h are defined on $\mathbb{R}$, the set of real numbers, as follows: $g : x \mapsto x^2 + 2$ and $h : x \mapsto x + 2$. Find:

a. Find $g(2)$.

b. Find $h^{-1}(x)$.

c. Find $hg(x)$.

Exercise 6 (5 marks)

Given that y varies directly as x and that $y = 10$ when $x = \frac{1}{3}$:

a. Find the relation between x and y.

b. Hence find the value of x when $y = 6$.

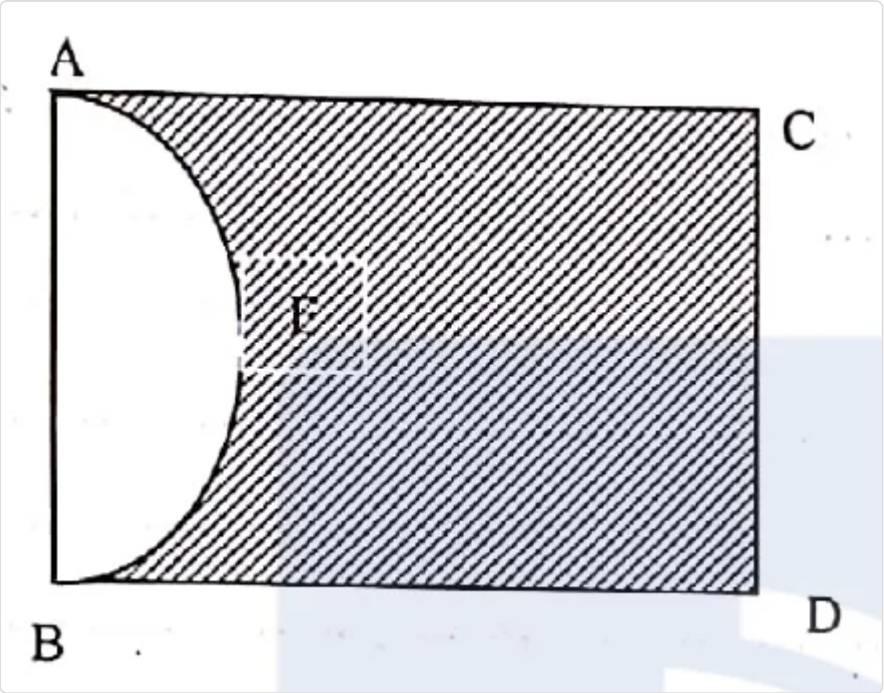
Exercise 7 (4 marks)

The angle of elevation of the top of a tower from a point M, $20\,\text{m}$ away from the base of the tower on a level ground, is $050°$. Find:

a. Find the height of the tower to one decimal place.

b. Find the angle of depression of the point M from the top of the tower.

Exercise 8 (6 marks)



In Figure 2, AEB is a semi-circle. $ABCD$ is a rectangle with $AC = 20\,\text{m}$ and $AB = 14\,\text{m}$. Find the area of the shaded region. *[6 marks]*

Exercise 9 (5 marks)

a. The sum of the first n terms of a sequence is given by $S_n = 2n^2 - n, n \in \mathbb{Z}^*$. Find the first three terms of the sequence.

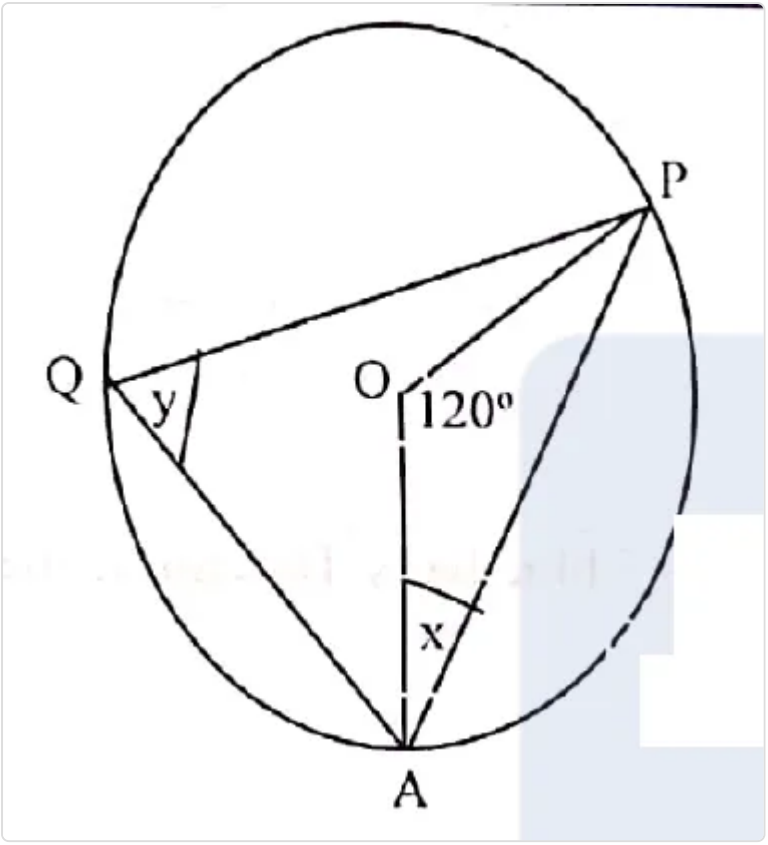
b. State the type of sequence.

Exercise 10 (5 marks)

a. Given the matrices $A = \begin{pmatrix} 5 & 2 \\ 3 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 0 \\ 1 & 4 \end{pmatrix}$, state the transpose of the matrix A.

b. Given the matrices $A = \begin{pmatrix} 5 & 2 \\ 3 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 0 \\ 1 & 4 \end{pmatrix}$, find the matrix $A - B$.

Exercise 11 (5 marks)



In Figure 3, APQ is a circum-circle. The chord AP subtends an angle of 120° at the centre, O, of the circle. Calculate the values of the angles:

a. Calculate the value of the angle x.

b. Calculate the value of the angle y.

Exercise 12 (6 marks)

The coordinates of the points A, B, C are $(1, 3)$, $(2, y)$ and $(4, 6)$ respectively. Given that A, B and C are on a straight line, find:

a. Find the value of y.

b. Find the length of the line segment AB, leaving your answer in surd form.

c. Find the ratio $AB : BC$.

Exercise 13 (6 marks)

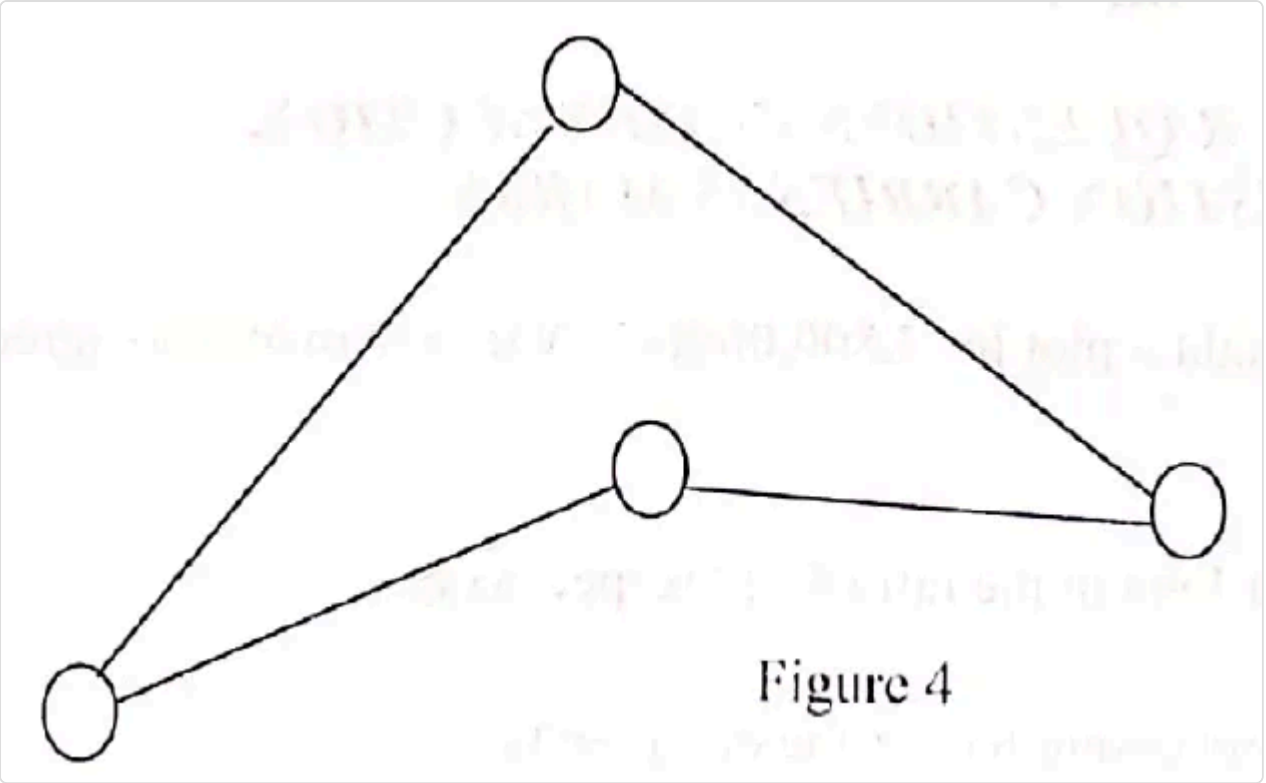
A ball is drawn at random from a box containing 6 red balls, 4 white balls and 5 blue balls. Determine the probability that it is:

a. A white ball.

b. Not a red ball.

c. A red or white ball.

Exercise 14 (4 marks)



In the network graph in Figure 4, state:

a. State the number of nodes.

b. State the number of edges.

c. State the number of regions.

Exercise 15 (5 marks)

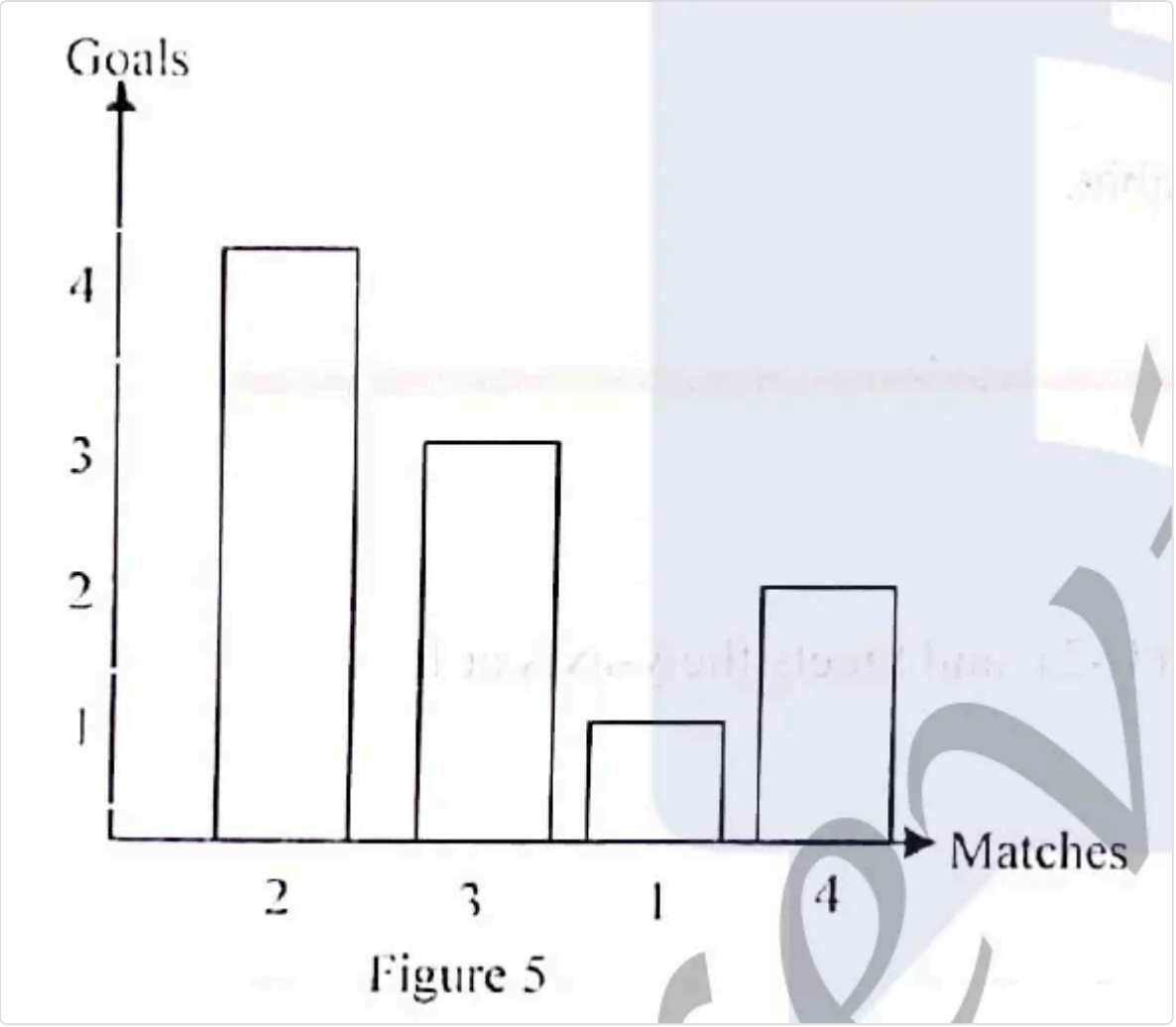


Figure 5

Figure 5 is a bar chart showing 10 matches played and the number of goals scored per match by a certain team last football season. Find:

a. Find the number of goals scored in two matches by the team.

b. Find the total number of goals scored by the team last football season.

---

ia. Three friends, Michael, Peter and John, bought a plot for $2{,}800{,}000$ FCFA from a man who agreed to be paid 65% of its value as initial deposit. Calculate the initial deposit.

ib. The deposit is paid by Michael, Peter and John in the ratio $5:3:2$ respectively. Determine the amount paid by Peter.

ic. Given that $91\ 000$ FCFA is to be paid as processing fee for the documents, find the percentage of the initial deposit needed to process the documents.

id. Calculate the balance expected to be paid to the man.

iia. Given the matrix $M = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$, find the determinant of M.

iib. Given the matrix $M = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$, find the adjugate of M.

iic. Hence find the inverse of M.

ia. The scores of twenty students in a Physics test are recorded as follows:

$70 \ \ 80 \ \ 78 \ \ 98 \ \ 84 \ \ 67 \ \ 98 \ \ 70 \ \ 80 \ \ 100$

$87 \ \ 83 \ \ 70 \ \ 70 \ \ 88 \ \ 91 \ \ 70 \ \ 78 \ \ 88 \ \ 88$

Represent the above scores on a frequency distribution table.

ib. Determine the mode of the distribution.

ic. Determine the median score.

id. Determine the mean score.

iia. A straight line l_1 passes through the points $P(3, 4)$ and $Q(1, -2)$, and meets the y-axis at R. Determine the equation of the line l_1.

iib. A straight line l_1 passes through the points $P(3, 4)$ and $Q(1, -2)$, and meets the y-axis at R. Find the coordinates of the point R.

---

a. Given the coordinates of the vertices of triangle ABC as $A(1, 4)$, $B(1, 1)$ and $C(3, 1)$, find the coordinates of the vertices of the triangle $A'B'C'$ obtained by rotating triangle ABC through $90°$, anticlockwise, about the origin.

b. On a graph paper, taking 1 cm to represent 1 unit on both axes for values $-4 \leq x \leq 7$ and $-2 \leq y \leq 10$, plot triangle ABC and $A'B'C'$.

c. Draw the line $x + y = 6$.

d. Triangle $A'B'C'$ is mapped onto triangle $A''B''C''$ by a reflection in the mirror line $x + y = 6$. Reflect $A'B'C'$ in the line $x + y = 6$.

e. Determine the coordinates of triangle $A''B''C''$.

---

ia. Given the function $f(x) = 5 + 3x - x^2$, taking values of x from -2 to $+5$ and using a scale of 1 cm to represent 1 unit on both axes, draw the graph of $f(x)$.

ib. Using your graph, determine the values of x for which $5 + 3x - x^2 = 0$.

ic. Find the gradient of the curve at the point where it cuts the y-axis.

iia. P and Q have position vectors $3\mathbf{i} + 2\mathbf{j}$ and $-\mathbf{i} + 2\mathbf{j}$ respectively, and $\overrightarrow{OP} = 3\overrightarrow{OQ} + 2\overrightarrow{OR}$. Find $2\overrightarrow{OR}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

iib. Find the position vector of R.

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.